# Pancyclicity of $[p+2, p]$-graphs *

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**Abstract**

An $[s,t]$-graph is a graph in which every induced subgraph on $s$ vertices contains at least $t$ edges. This concept originates from the classical Chvátal-Erdős condition for Hamiltonicity. Zhan conjectured that, for $p \in \{3,4,5\}$, every 2-connected $[p+2,p]$-graph of order at least $2p+3$ and minimum degree at least $p$ is pancyclic. Liu and Liu recently proved this conjecture for $p=3$. In this paper, we prove that every 2-connected $[6,4]$-graph of order at least 153 and minimum degree at least 4 is pancyclic, thereby confirming the $p=4$ case of Zhan's conjecture for graphs of orders at least 153. We also construct an infinite family of counterexamples to the $p=5$ case of the conjecture. As a preliminary result, we show that every 2-connected $[6,4]$-graph of order at least 10 and minimum degree at least 3 is hamiltonian. To establish pancyclicity, we combine this hamiltonicity theorem with a path-shortening proposition of Draganić, Munhá Correia and Sudakov, together with results on cycle-complete Ramsey numbers.

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## 1 Introduction

A Hamilton path in a graph $G$ is a path containing all vertices of $G$, and a Hamilton cycle in $G$ is a cycle containing all vertices of $G$. A graph is hamiltonian if it contains a Hamilton cycle. The hamiltonian problem is one of the central problems in graph theory: although deciding hamiltonicity is NP-complete, many classical and modern theorems give useful sufficient conditions. Dirac's theorem, Ore's theorem, and the Chvátal–Erdős theorem are among the standard examples; see [6, 14, 26] for general background.

Pancyclicity is a strengthening of hamiltonicity. A graph of order $n$ is pancyclic if it contains a cycle of length $k$ for every integer $k$ with $3 \le k \le n$. Thus every pancyclic graph

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is hamiltonian. Since Bondy's work on pancyclic graphs, a recurring theme has been that many nontrivial hamiltonian conditions also force pancyclicity, perhaps with a small family of exceptional graphs [3, 4].

The notion of an $[s,t]$-graph provides a local density analogue of the independence number. A graph $G$ is called an $[s,t]$-graph if every induced subgraph of $G$ on $s$ vertices contains at least $t$ edges. This definition extends the condition on independence number, since $\alpha(G) \leq k$ is equivalent to saying that $G$ is a $[k+1,1]$-graph. Properties of $[s,t]$-graphs related to hamiltonicity and pancyclicity have attracted increasing attention. In 1972, Chvátal and Erdős [9] proved the following result.

**Theorem 1.1** (Chvátal and Erdős [9]). *Let $G$ be a $k$-connected graph. If $G$ is a $[k+1,1]$-graph, then $G$ is Hamiltonian.*

The pancyclicity of graphs satisfying Chvátal-Erdős-type conditions has been extensively studied in graph theory. In particular, Jackson and Ordaz [15] conjectured in 1990 that every $k$-connected $[k,1]$-graph is pancyclic. Recently, Draganić, Munhá Correia and Sudakov [12] resolved the conjecture asymptotically. Inspired by this conjecture, we investigate the pancyclicity of $[s,t]$-graphs. Note that every $[s,t]$-graph is also an $[s+1,t+1]$-graph. Thus, it is natural to study $[k+2,2]$-graphs as a further extension of the Chvátal–Erdős condition.

In 2005, Liu and Wang [21] proved that every 2-connected $[4,2]$-graph of order at least six is hamiltonian. Later, in 2008, Liu [19] strengthened this direction by proving that every 2-connected $[4,2]$-graph of order at least seven is pancyclic. An alternative proof was presented in [28]. Recent developments on $[4,2]$-graphs can be found in [8, 18]. For $[5,3]$-graphs, Li and Wang established the following hamiltonicity theorem.

**Theorem 1.2** (Li and Wang [17]). *Every 2-connected $[5,3]$-graph of order at least eight with minimum degree at least three is hamiltonian.*

Motivated by the above results, Zhan [29] proposed the following pancyclicity conjecture for $[p+2,p]$-graphs.

**Conjecture 1.3** (Zhan [29]). *Let $p \in \{3,4,5\}$. Every 2-connected $[p+2,p]$-graph of order at least $2p+3$ and with minimum degree at least $p$ is pancyclic.*

Recently, Liu and Liu verified the case $p = 3$.

**Theorem 1.4** (Liu and Liu [20]). *Every 2-connected $[5,3]$-graph of order at least nine with minimum degree at least three is pancyclic.*

For $p = 5$, Theorem 3.1 below constructs an infinite family of 2-connected $[7,5]$-graphs with minimum degree at least 5 that are not hamiltonian. Therefore the case $p = 5$ of Conjecture 1.3 does not hold.

We then turn to the remaining case $p = 4$, that is, to pancyclicity of $[6,4]$-graphs. Zhan's conjecture predicts that every 2-connected $[6,4]$-graph of order at least 11 and minimum degree at least 4 is pancyclic. Our main pancyclic result for $[6,4]$-graphs verifies this statement for orders at least 153.

**Theorem 1.5.** *Every 2-connected $[6,4]$-graph of order at least 153 with minimum degree at least 4 is pancyclic.*

For the general case, we prove the pancyclicity of $G$ in two structural settings: when $\alpha(G) \leq 2$ and when $\alpha(G) = 5$. As a preliminary result for Theorem 1.5, we prove the following hamiltonicity theorem for $[6,4]$-graphs.

**Theorem 1.6.** *Every 2-connected $[6,4]$-graph of order at least ten with minimum degree at least three is hamiltonian.*

Since every $[5,3]$-graph is a $[6,4]$-graph, Theorem 1.6 extends Li–Wang's hamiltonicity theorem for 2-connected $[5,3]$-graphs of order at least eight with minimum degree at least three [17] to a wider local-density class in the range $n \geq 10$. It also works under the weaker minimum-degree condition $\delta(G) \geq 3$ than the $\delta(G) \geq 4$ required in the $p = 4$ pancyclic conjecture. We remark that the lower bound in Theorem 1.6 is sharp.

**Remark 1.7.** *The graph $K_{4,5}$ is a non-hamiltonian 4-connected $[6,4]$-graph of order nine with minimum degree four. Hence the assumption $n \geq 10$ in Theorem 1.6 cannot be weakened.*

The paper is organized as follows. Section 2 collects notation and auxiliary results. Section 3 gives the construction for the case $p = 5$. Section 4 proves the hamiltonicity theorem, Theorem 1.6. Section 5 proves the pancyclicity results for $[6,4]$-graphs, including Theorem 1.5. Section 6 contains concluding remarks and records a pancyclic edge version suggested by Zhan.

## 2 Preliminaries

We consider finite simple graphs. For a graph $G$, let $V(G)$ and $E(G)$ denote its vertex set and edge set. For $X \subseteq V(G)$, let $G[X]$ be the subgraph induced by $X$, and let $e(X) = |E(G[X])|$. For a vertex $v$ and a subgraph $H$ of $G$, write

$$N_H(v) = N_G(v) \cap V(H).$$

For a vertex set $X$, put $N_H(X) = \bigcup_{x \in X} N_H(x)$. If $P$ is an oriented path and $v \in V(P)$, then $v^+$ and $v^-$ denote the successor and predecessor of $v$ on $P$, whenever they exist. For two vertices $a, b \in V(P)$, write $P[a,b]$ for the segment of $P$ between $a$ and $b$. If the orientation matters, $aPb$ denotes the segment from $a$ to $b$ following the orientation of $P$, while $a\overleftarrow{P}b$ denotes the same segment in the reverse direction.

We shall also use the following notion of toughness. For a connected noncomplete graph $G$, its toughness is

$$\tau(G) = \min\left\{\frac{|S|}{\omega(G-S)} : S \subseteq V(G),\ \omega(G-S) \geq 2\right\},$$

where $\omega(G-S)$ denotes the number of components of $G-S$; for a complete graph, set $\tau(G) = \infty$. A graph is 1-tough if $\tau(G) \geq 1$, or equivalently, if $\omega(G-S) \leq |S|$ for every cutset $S$ with $\omega(G-S) \geq 2$. Every hamiltonian graph is 1-tough.

The $n$-closure $\mathrm{cl}(G)$ of a graph $G$ of order $n$ is obtained by repeatedly adding a non-edge $uv$ with $d(u) + d(v) \geq n$ until no such non-edge remains. We shall use the following theorem of Bondy and Chvátal.

**Theorem 2.1** (Bondy and Chvátal [5])**.** *A graph $G$ is hamiltonian if and only if its closure $\mathrm{cl}(G)$ is hamiltonian.*

In particular, if $G$ is closed and non-hamiltonian, then

$$uv \notin E(G) \quad \Longrightarrow \quad d(u) + d(v) \leq n - 1. \tag{1}$$

We also need the following three hamiltonicity results involving 1-tough graphs.

**Theorem 2.2** (Bigalke and Jung [2]). *If $G$ is a 1-tough non-hamiltonian graph with $\kappa(G) \geq 3$, then either $G$ is the Petersen graph or $\alpha(G) \geq \kappa(G) + 2$.*

**Theorem 2.3** (Jung [16]). *Every 1-tough graph $G$ of order $n \geq 11$ with $\delta(G) \geq \frac{n-4}{2}$ is hamiltonian.*

**Theorem 2.4** (Moon and Moser [22]). *Let $G$ be a balanced bipartite graph with partite sets of size $m$. If $\delta(G) \geq \frac{m+1}{2}$, then $G$ has a Hamilton cycle.*

## 3 A construction for the case $p = 5$

In this section we show that the case $p = 5$ of Conjecture 1.3 does not hold. Recall that the join $G \vee H$ of two vertex-disjoint graphs $G$ and $H$ is obtained from their disjoint union by adding all edges between $V(G)$ and $V(H)$.

For an integer $t \geq 4$, let

$$F_t = K_2 \vee 3K_t,$$

where $3K_t$ denotes the disjoint union of three copies of $K_t$. Let $x, y$ be the two vertices of the $K_2$ and let $C_1, C_2, C_3$ be the three copies of $K_t$.

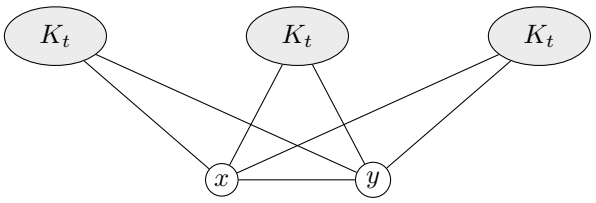



Figure 1: $F_t = K_2 \vee 3K_t$. The three grey ellipses, arranged in one row, represent the three copies of $K_t$; both $x$ and $y$ are adjacent to every vertex in each copy.

**Theorem 3.1.** *For every integer $t \geq 4$, the graph $F_t = K_2 \vee 3K_t$ is a 2-connected $[7, 5]$-graph with $\delta(F_t) = t + 1 \geq 5$, but $F_t$ is not 1-tough and hence is not hamiltonian. In particular, it is not pancyclic; consequently, the case $p = 5$ of Conjecture 1.3 does not hold.*

*Proof.* Clearly $|V(F_t)| = 3t + 2 \geq 14$ and $\delta(F_t) = t + 1 \geq 5$. The graph $F_t$ is 2-connected, since after deleting any one vertex, at least one of $x$ and $y$ remains and is adjacent to every vertex outside $\{x, y\}$.

It remains only to check the $[7, 5]$ condition. Let $S \subseteq V(F_t)$ with $|S| = 7$. If $S$ contains $x$ or $y$, then $S$ induces at least six edges. Otherwise, writing $a_i = |S \cap V(C_i)|$, we have $a_1 + a_2 + a_3 = 7$ and

$$e(F_t[S]) = \binom{a_1}{2} + \binom{a_2}{2} + \binom{a_3}{2} \geq 5,$$

with equality possible for the distribution $3, 2, 2$. Thus $F_t$ is a $[7, 5]$-graph.

Finally, $F_t - \{x, y\}$ has three components, and so $F_t$ is not 1-tough. Hence $F_t$ is not hamiltonian, and therefore not pancyclic. □

**Remark 3.2.** *In [28], Zhan proved that, for even $p \geq 6$, there exist $[p+2, p]$-graphs that are not pancyclic by constructing a family of triangle-free graphs. When $p \geq 5$ and $p+2 \leq 3t+2$, one can verify that $F_t$ is a $[p+2, p]$-graph. However, $F_t$ is not hamiltonian. This shows that there exist non-pancyclic $[p+2, p]$-graphs for all $p \geq 5$.*

## 4 A preliminary hamiltonian result: proof of Theorem 1.6

Let $G$ be a 2-connected $[6, 4]$-graph of order $n \geq 10$ with $\delta(G) \geq 3$. We shall prove that $G$ is hamiltonian. The proof is divided into the 3-connected case and the 2-cut case, but the argument is presented through claims.

**Claim 1.** *If $G$ is 3-connected, then $G$ is 1-tough.*

*Proof.* Suppose not. Then there is a cutset $S \subseteq V(G)$ such that

$$c := \omega(G - S) > |S|.$$

Since $G$ is 3-connected, $|S| \geq 3$. If $c \geq 6$, then choosing one vertex from each of six components of $G - S$ gives six independent vertices, contradicting the $[6, 4]$ condition. Hence $c \leq 5$.

Thus the only possibilities are

$$(c, |S|) \in \{(4, 3), (5, 3), (5, 4)\}.$$

If $c = 5$, then since $n \geq 10$, at least one component of $G - S$ has order at least two. Taking two vertices from this component and one vertex from each of the other four components gives six vertices inducing at most one edge, a contradiction.

Now let $c = 4$. Then $|S| = 3$. If two components of $G - S$ have order at least two, then choosing two vertices from each of these two components and one vertex from each of the remaining two components gives six vertices inducing at most two edges, impossible. Therefore at most one component is not a singleton. Since

$$|V(G - S)| = n - |S| \geq 7,$$

that component has order at least four. Taking three vertices from it and the three singleton components gives six vertices inducing at most $\binom{3}{2} = 3$ edges, again impossible. Hence $G$ is 1-tough. □

**Claim 2.** *If $G$ is 3-connected, then $G$ is hamiltonian.*

*Proof.* Suppose, to the contrary, that $G$ is not hamiltonian. Let $H = \mathrm{cl}(G)$ be the Bondy–Chvátal closure of $G$. Adding edges preserves 3-connectivity and the $[6, 4]$ property. By Theorem 2.1, the graph $H$ is still non-hamiltonian. Thus $H$ is closed, and (1) holds in $H$. By Claim 1, $H$ is 1-tough.

Since $H$ is a $[6, 4]$-graph, it has no independent set of order six. Hence

$$\alpha(H) \leq 5.$$

Suppose first that $\alpha(H) \leq 4$. By Theorem 2.2, the graph $H$ must be the Petersen graph, because otherwise

$$\alpha(H) \geq \kappa(H) + 2 \geq 5.$$

But the Petersen graph is not a $[6,4]$-graph. Indeed, with the usual notation $u_i, v_i$ for $i \in \mathbb{Z}_5$, where the edges are $u_iu_{i+1}$, $u_iv_i$, and $v_iv_{i+2}$, the set

$$I = \{u_0, u_2, v_3, v_4\}$$

is independent. The Petersen graph has 15 edges and each vertex of $I$ has degree three, so the six vertices in the complement of $I$ induce $15 - 4 \cdot 3 = 3$ edges, contrary to the $[6,4]$ property.

Thus we may assume that

$$\alpha(H) = 5.$$

Let $I$ be an independent set of order five. For every $v \in V(H) \setminus I$, the six vertices $I \cup \{v\}$ have all their induced edges incident with $v$, and therefore

$$d_I(v) \geq 4.$$

Consequently

$$\sum_{x \in I} d_H(x) = e_H(I, V(H) \setminus I) \geq 4(n-5). \tag{2}$$

On the other hand, every pair of vertices of $I$ is a non-edge of the closed graph $H$. Hence, by (1),

$$d_H(x) + d_H(y) \leq n - 1 \qquad (x, y \in I,\ x \neq y).$$

Summing over all pairs in $I$ gives

$$4\sum_{x \in I} d_H(x) = \sum_{\{x,y\} \subseteq I} \big(d_H(x) + d_H(y)\big) \leq \binom{5}{2}(n-1). \tag{3}$$

Combining (2) and (3), we obtain

$$4(n-5) \leq \frac{5}{2}(n-1),$$

and so $n \leq 11$.

Assume first that $n = 11$. Let $O = V(H) \setminus I$. Then $|O| = 6$, and $d_I(v) \geq 4$ for every $v \in O$. Hence

$$\sum_{x \in I} d_H(x) = e_H(I, O) \geq 24.$$

If some $x \in I$ had $d_H(x) \leq 3$, then every pair in $I \setminus \{x\}$ would have degree-sum at most ten by (1). Therefore

$$3 \sum_{y \in I \setminus \{x\}} d_H(y) = \sum_{\{y,z\} \subseteq I \setminus \{x\}} \big(d_H(y) + d_H(z)\big) \leq 6 \cdot 10,$$

and hence $\sum_{y \in I \setminus \{x\}} d_H(y) \leq 20$. This would give $\sum_{y \in I} d_H(y) \leq 23$, a contradiction. Thus every vertex of $I$ has degree at least four. Every vertex of $O$ has at least four neighbours in $I$, and hence also has degree at least four. Thus

$$\delta(H) \geq 4 \geq \frac{11-4}{2}.$$

By Theorem 2.3, the graph $H$ is hamiltonian, a contradiction.

It remains to consider $n = 10$. Again let $O = V(H) \setminus I$. Then $|I| = |O| = 5$. Consider the bipartite graph

$$B = H[I, O].$$

For every $v \in O$, we have $d_B(v) = d_I(v) \geq 4$. For every $x \in I$, since $I$ is independent and $V(H) = I \cup O$, we have

$$d_B(x) = d_H(x) \geq \delta(H) \geq 3.$$

Thus $B$ is a balanced bipartite graph with partite sets of size five and $\delta(B) \geq 3 = (5+1)/2$. By Theorem 2.4, the graph $B$ has a Hamilton cycle. This cycle is also a Hamilton cycle of $H$, a contradiction.

Therefore $H$, and hence $G$, is hamiltonian. □

We may now assume that $G$ is not 3-connected. Since $G$ is 2-connected, it has a 2-cut. The following claim records elementary consequences of the $[6,4]$ condition across such a cut.

**Claim 3.** *Let $\{x,y\}$ be a 2-cut of $G$. Then $G - \{x,y\}$ has exactly two components, say $A$ and $B$. Moreover:*

(a) $|A|, |B| \geq 2$;

(b) *every four vertices in $A$ induce at least three edges, and the same holds in $B$;*

(c) *if one component contains an independent set of order three, then the other component has order two;*

(d) *if $B = \{r,s\}$, then $rs \in E(G)$ and $\{r,s\}$ is complete to $\{x,y\}$.*

*Proof.* No component of $G - \{x,y\}$ is a singleton, since such a vertex would have degree at most two, contrary to $\delta(G) \geq 3$. Thus every component has order at least two.

If $G - \{x,y\}$ had at least three components, then choosing two vertices from each of three components would give six vertices inducing at most three edges, contradicting the $[6,4]$ property. Hence $G - \{x,y\}$ has exactly two components, and (a) follows.

For (b), take four vertices in one component and two vertices in the other. There are no edges between the two components, and the two vertices in the other component contribute at most one edge. Therefore the four vertices in the first component must induce at least three edges.

For (c), suppose, say, that $A$ contains an independent triple. If $|B| \geq 3$, then this triple together with any three vertices of $B$ would induce at most three edges, impossible. Thus $|B| = 2$ by (a).

Finally, suppose $B = \{r,s\}$. Since $B$ is connected, $rs \in E(G)$. The vertex $r$ has no neighbours outside $\{r,s,x,y\}$, and $\delta(G) \geq 3$; hence $r$ is adjacent to both $x$ and $y$. The same argument applies to $s$. □

The next claim is the bridge lemma for a 2-cut. It is the only place where we use a longest path inside one side of the cut.

**Claim 4.** *Let $\{x,y\}$ be a 2-cut of $G$, and let $X$ be one component of $G - \{x,y\}$. Then $G[X \cup \{x,y\}]$ has a Hamilton path with endpoints $x$ and $y$.*

*Proof.* Let $Y$ be the other component of $G - \{x,y\}$, and put

$$H = G[X \cup \{x,y\}].$$

Since $G$ is 2-connected, the component $X$ has a neighbour of $x$ and a neighbour of $y$; otherwise one of $x, y$ would be a cut vertex. Thus $H$ contains an $x$-$y$ path.

We first dispose of the small cases. If $|X| = 2$, then Claim 3(d) gives a Hamilton $x$-$y$ path through the two vertices of $X$. Let $|X| = 3$, say $X = \{a, b, c\}$. Since $X$ is connected, either $G[X]$ is a path or $G[X] = K_3$. If $G[X]$ is a path, say $abc$, then the endpoints $a$ and $c$ have degree one inside $X$. Their only possible neighbours outside $X$ are $x$ and $y$, and so $\delta(G) \geq 3$ forces both $a$ and $c$ to be adjacent to both $x$ and $y$. Thus $xabcy$ is a Hamilton $x$-$y$ path in $H$. If $G[X] = K_3$, then $N_X(x)$ and $N_X(y)$ are nonempty. We can choose distinct vertices $p \in N_X(x)$ and $q \in N_X(y)$; otherwise both neighbourhoods would be the same singleton, leaving the other two vertices of $X$ with degree two. Since $G[X] = K_3$, there is a Hamilton path of $G[X]$ from $p$ to $q$, and this extends to a Hamilton $x$-$y$ path in $H$.

Henceforth assume that $|X| \geq 4$. Let

$$P = v_0 v_1 \cdots v_\ell$$

be a longest $x$-$y$ path in $H$, where $v_0 = x$ and $v_\ell = y$. We shall show that $P$ covers $X$. Suppose not. Then $X \setminus V(P)$ is nonempty.

**Claim 5.** *Every component of $X \setminus V(P)$ is a singleton.*

*Proof.* Let $D$ be a component of $X \setminus V(P)$. The component $D$ has at least one neighbour on $P$; otherwise $D$ would be separated from the rest of $G$. In fact, $D$ has at least two neighbours on $P$. Indeed, $D$ has no neighbours in $Y$, and it has no neighbours in any other component of $X \setminus V(P)$. Thus, if $D$ had a unique neighbour $z$ on $P$, then $G - z$ would disconnect $D$ from the rest of the graph, contradicting the 2-connectivity of $G$.

Suppose that $|D| \geq 2$. Among all paths whose two ends are distinct vertices of $N_P(D)$ and whose internal vertices lie in $D$, choose one, say $Q$, with as many internal vertices as possible. Such a path exists because $D$ is connected and has at least two neighbours on $P$.

We first note that $Q$ contains at least two vertices of $D$ as internal vertices. Suppose instead that $Q$ has only one internal vertex, say $d$. If some vertex $d' \in D \setminus \{d\}$ also has a neighbour on $P$, then a path in $D$ from $d$ to $d'$, together with one of the two $P$-neighbours of $d$ used by $Q$ and a suitable $P$-neighbour of $d'$, gives a path of the same type using more vertices of $D$, a contradiction to the choice of $Q$. Hence $d$ would be the only vertex of $D$ adjacent to $P$. Since $|D| \geq 2$, deleting $d$ would then separate a component of $D - d$ from the rest of $G$, again contradicting the 2-connectivity of $G$.

Let $u$ and $v$ be the ends of $Q$, ordered so that $u$ precedes $v$ on $P$. By the maximality of $P$, the segment $P[u, v]$ contains at least as many internal vertices as $Q$; otherwise replacing $P[u, v]$ with $Q$ would give a longer $x$-$y$ path. Since $Q$ contains at least two vertices of $D$, it follows that $P[u, v]$ has at least two internal vertices. Thus $u^+$ and $v^-$ are distinct vertices of $X$.

Neither $u^+$ nor $v^-$ has a neighbour in $D$. Indeed, if $u^+$ had a neighbour in $D$, then, since $u$ also has a neighbour in $D$ and $D$ is connected, the edge $uu^+$ of $P$ could be replaced by a path from $u$ to $u^+$ whose internal vertices lie in $D$, producing a longer $x$-$y$ path. The proof for $v^-$ is symmetric.

Choose two distinct vertices $d_1, d_2 \in D$. The four vertices

$$\{u^+, v^-, d_1, d_2\} \subseteq X$$

induce at most two edges: possibly $u^+v^-$ and possibly $d_1d_2$. This contradicts Claim 3(b). Therefore $|D| = 1$. □

Let $\{w\}$ be a component of $X \setminus V(P)$.

**Claim 6.** *The vertex $w$ has exactly three neighbours on $P$. Moreover, if*

$$S_w = \{p^+ : p \in N_P(w) \text{ and } p^+ \in X\},$$

*then $S_w$ is independent and nonadjacent to $w$.*

*Proof.* Since $w$ is a singleton component of $X \setminus V(P)$, it has no neighbours in $X \setminus V(P)$. It also has no neighbours in $Y$. Thus all neighbours of $w$ lie on $P$, and

$$|N_P(w)| \geq 3. \tag{4}$$

No two neighbours of $w$ are consecutive on $P$, because otherwise $w$ could be inserted between them to obtain a longer $x$-$y$ path. It follows in particular that $w$ is anticomplete to $S_w$.

Now take two vertices $p_i, p_j \in N_P(w)$, with $p_i$ preceding $p_j$ on $P$, and assume $p_i^+, p_j^+ \in X$. If $p_i^+ p_j^+ \in E(G)$, then

$$xPp_i\, w\, p_j \overleftarrow{P} p_i^+\, p_j^+ Py$$

is an $x$-$y$ path longer than $P$, a contradiction. Hence $S_w$ is independent.

At most one neighbour of $w$ fails to contribute a vertex to $S_w$. To see this, note that a neighbour $p$ of $w$ on $P$ has $p^+ \notin X$ only in the following two cases: either $p = y$, which has no successor, or $p = y^-$, whose successor is $y$. The vertices $y^-$ and $y$ are consecutive on $P$, and $w$ is adjacent to at most one of two consecutive vertices. Therefore at most one neighbour of $w$ has no successor in $X$.

If $|N_P(w)| \geq 4$, then the preceding paragraph gives $|S_w| \geq 3$. Any three vertices of $S_w$, together with $w$, would then be four vertices in $X$ inducing no edge, contrary to Claim 3(b). Combining this with (4), we obtain $|N_P(w)| = 3$. □

**Claim 7.** *There is exactly one vertex of $X$ outside $P$. Moreover, $|X| = 6$, and after possibly reversing $P$ and relabelling vertices,*

$$P = xp_1q_1p_2q_2p_3y, \qquad N_P(w) = \{p_1, p_2, p_3\},$$

*where $X \setminus V(P) = \{w\}$.*

*Proof.* By Claim 6, at most one neighbour of $w$ fails to contribute a vertex to $S_w$, and $|N_P(w)| = 3$. Hence $|S_w| \geq 2$. Choose two vertices $q_1, q_2 \in S_w$. The set $\{w, q_1, q_2\}$ is independent by Claim 6. Claim 3(c) therefore implies that the other component has order two; write

$$Y = \{r, s\}.$$

Since $n \geq 10$, we have

$$|X| = n - 2 - |Y| \geq 6. \tag{5}$$

There cannot be a second component of $X \setminus V(P)$. If $\{w'\}$ were another such singleton, then the four vertices $\{w, w', q_1, q_2\}$ would induce at most two edges in $X$: there is no edge $ww'$, the vertices $w, q_1, q_2$ are independent, and only the two edges from $w'$ to $q_1, q_2$ could occur. This contradicts Claim 3(b). Thus

$$X \setminus V(P) = \{w\}.$$

Let

$$M = (X \cap V(P)) \setminus N_P(w).$$

The vertices $q_1$ and $q_2$ belong to $M$. If $M$ contained a third vertex $q_3$, then $\{w, q_1, q_2, q_3\}$ would induce at most two edges in $X$, again contradicting Claim 3(b). Therefore

$$M = \{q_1, q_2\}. \tag{6}$$

Since $|N_P(w)| = 3$, equation (6) gives

$$|X| = 1 + |M| + |N_P(w) \cap X| \leq 1 + 2 + 3 = 6.$$

Together with (5), this implies that $|X| = 6$ and that all three neighbours of $w$ on $P$ lie in $X$.

The vertices of $X \cap V(P)$ consist of the three neighbours of $w$ and the two vertices $q_1, q_2$. No two neighbours of $w$ are consecutive on $P$. Hence these five vertices occur on $P$ in the alternating order

$$p_1, q_1, p_2, q_2, p_3$$

after possibly reversing $P$ and relabelling them. This gives the claimed form of $P$. □

**Claim 8.** *The configuration in Claim 7 cannot occur.*

*Proof.* For each $i \in \{1, 2, 3\}$, the four vertices

$$\{w, q_1, q_2, p_i\}$$

lie in $X$ and therefore induce at least three edges by Claim 3(b). In this four-set, the edge $wp_i$ is present, while $wq_1$, $wq_2$, and $q_1q_2$ are absent. Hence

$$p_iq_1,\ p_iq_2 \in E(G) \qquad (i = 1, 2, 3). \tag{7}$$

Since $Y = \{r, s\}$, Claim 3(d) gives

$$rs,\ rx,\ sx \in E(G).$$

Consider the six vertices

$$T = \{q_1, q_2, w, r, s, x\}.$$

There are no edges from $\{r, s\}$ to $X$, the set $\{w, q_1, q_2\}$ is independent, and $xw \notin E(G)$ because all three neighbours of $w$ on $P$ lie in $X$. Thus, apart from the possible edges $xq_1$ and $xq_2$, the set $T$ induces at most the three edges $rs$, $rx$, and $sx$. The $[6, 4]$ property therefore gives

$$xq_1 \in E(G) \quad \text{or} \quad xq_2 \in E(G). \tag{8}$$

If $xq_1 \in E(G)$, then by (7) the sequence

$$xq_1p_2wp_1q_2p_3y$$

is a Hamilton $x$-$y$ path of $H$. If $xq_2 \in E(G)$, then

$$xq_2p_2wp_1q_1p_3y$$

is a Hamilton $x$-$y$ path of $H$. In either case we obtain an $x$-$y$ path in $H$ longer than $P$, contradicting the choice of $P$. □

Claim 8 contradicts the assumption that $X \setminus V(P)$ is nonempty. Hence $P$ covers all vertices of $X$, and $H = G[X \cup \{x, y\}]$ has a Hamilton path from $x$ to $y$. □

We now finish the proof of Theorem 1.6. If $G$ is 3-connected, the result follows from Claim 2. Thus assume that $G$ is not 3-connected, and choose a 2-cut $\{x, y\}$. By Claim 3, the graph $G - \{x, y\}$ has exactly two components, say $A$ and $B$. By Claim 4, the graph $G[A \cup \{x, y\}]$ has a Hamilton path $P_A$ from $x$ to $y$, and $G[B \cup \{x, y\}]$ has a Hamilton path $P_B$ from $x$ to $y$. The paths $P_A$ and $P_B$ are internally disjoint and have the same endpoints, so their union is a Hamilton cycle of $G$. This completes the proof of Theorem 1.6.

## 5 Pancyclicity of $[6, 4]$-graphs

We now prove pancyclicity results for the remaining $p = 4$ case of Conjecture 1.3. We first record a full small-independence case, namely $\alpha(G) \leq 2$ in the conjectured range. We then separate the case $\alpha(G) = 5$ from the case $\alpha(G) \leq 4$. The first case uses the local structure forced by the $[6, 4]$ condition, while the second case combines Theorem 1.6, a path-shortening proposition of Draganić, Munhá Correia and Sudakov, and cycle-complete Ramsey numbers.

We shall use the following theorem of Amar, Fournier and Germa on pancyclism in Chvátal–Erdős graphs.

**Theorem 5.1** (Amar, Fournier and Germa [1])**.** *Let $G$ be a graph with $\alpha(G) \leq \kappa(G)$. If $\alpha(G) = 2$ and $G$ is not isomorphic to $C_4$ or $C_5$, then $G$ is pancyclic.*

**Corollary 5.2.** *Let $G$ be a 2-connected $[6, 4]$-graph of order $n \geq 11$ with $\delta(G) \geq 4$ and $\alpha(G) \leq 2$. Then $G$ is pancyclic.*

*Proof.* If $\alpha(G) = 1$, then $G$ is complete and hence pancyclic. Thus we may assume that $\alpha(G) = 2$. Since $G$ is 2-connected, $\kappa(G) \geq 2 = \alpha(G)$. By Theorem 5.1, the graph $G$ is pancyclic unless $G \cong C_4$ or $G \cong C_5$. These exceptions are impossible because $n \geq 11$. Hence $G$ is pancyclic. □

We shall also use a mild edge version of the same shortening idea. Following Li and Zhan [18], if $G$ has order $n$ and $A \subseteq \{3, 4, \ldots, n\}$, then an edge $e$ of $G$ is called *$A$-cyclic* if $e$ lies in a $k$-cycle of $G$ for every $k \in A$. In particular, a *pancyclic edge* is a $\{3, 4, \ldots, n\}$-cyclic edge.

We first record Proposition 2.9 of Draganić, Munhá Correia and Sudakov, which will be used to obtain all sufficiently long cycle lengths [11].

**Proposition 5.3** (Draganić, Munhá Correia and Sudakov [11])**.** *Let $G$ be a graph with independence number at most $k$, and let $P$ be a path in $G$ with endpoints $x$ and $y$. Let $c \in \mathbb{N}$ and let $U \subseteq V(P)$ be such that*

$$c\left(\frac{|V(P)| - (4c-1)|U|}{2k} - 1\right) > k.$$

*Then there exists an $x$-$y$ path $P'$ satisfying*

$$U \subseteq V(P') \subseteq V(P)$$

*and*

$$|V(P)| - (4c-3) \leq |V(P')| < |V(P)|.$$

The following form is the direct consequence of Proposition 5.3 obtained by taking $c = 1$ and $U = \emptyset$ and then iterating the shortening step along cycles.

**Theorem 5.4** (Draganić, Munhá Correia and Sudakov [11]). *Let $G$ be a hamiltonian graph of order $n$ with $\alpha(G) \le k$. Then $G$ contains a cycle of length $\ell$ for every integer $\ell$ with*

$$2k^2 + 2k \le \ell \le n.$$

*Moreover, if $C$ is a Hamilton cycle of $G$, then every edge of $C$ is $\{2k^2+2k, 2k^2+2k+1, \dots, n\}$-cyclic.*

*Proof.* We prove the edge statement, which implies the cycle-existence statement. Let $C$ be a cycle of length $L > 2k^2 + 2k$ and let $xy$ be an edge of $C$. Let $P = C - xy$ be the spanning $x$-$y$ path of $C$. In Proposition 5.3, take $c = 1$ and $U = \emptyset$. The hypothesis becomes

$$\frac{|V(P)|}{2k} - 1 > k,$$

that is, $L > 2k^2 + 2k$. Hence there is an $x$-$y$ path $P'$ with

$$L - 1 \le |V(P')| < L.$$

Thus $|V(P')| = L-1$, and adding the same edge $xy$ gives a cycle of length $L-1$ containing $xy$. Starting with a Hamilton cycle and a fixed edge $xy$ on it, and repeating this shortening step, gives cycles containing $xy$ of all lengths from $n$ down to $2k^2+2k$. Since $xy$ was arbitrary on the Hamilton cycle, every edge of that Hamilton cycle is $\{2k^2+2k, 2k^2+2k+1, \dots, n\}$-cyclic. □

For graphs $H$ and $F$, let $r(H, F)$ denote the Ramsey number, that is, the least integer $N$ such that every graph of order $N$ contains a copy of $H$ or its complement contains a copy of $F$. Cycle-complete Ramsey numbers were initiated by Erdős, Faudree, Rousseau and Schelp [13]. Equivalently, a graph $G$ of order at least $r(C_\ell, K_s)$ with $\alpha(G) < s$ must contain a cycle $C_\ell$.

For our application only the case $s = 5$ is needed. We shall use the following upper bound together with the exact value for $K_5$ and cycles of length at least five.

**Theorem 5.5** (Erdős, Faudree, Rousseau and Schelp [13]; Bollobás, Jayawardene, Yang, Huang, Rousseau and Zhang [7]). *For $\ell \ge 3$ and $s \ge 2$,*

$$r(C_\ell, K_s) \le \big((\ell-2)(s^{1/x}+2)+1\big)(s-1), \qquad x = \left\lfloor \frac{\ell-1}{2} \right\rfloor.$$

*Moreover, for every $\ell \ge 5$,*

$$r(C_\ell, K_5) = 4(\ell-1)+1.$$

**Corollary 5.6.** *Let $G$ be a 2-connected $[6,4]$-graph of order $n \ge 11$ with $\delta(G) \ge 4$ and $\alpha(G) = 5$. Then $G$ is pancyclic.*

*Proof.* Let $A = \{a_1, a_2, a_3, a_4, a_5\}$ be an independent set of order five, and put $B = V(G) \setminus A$. Write $m = |B|$. Then $m = n - 5 \ge 6$. Since $G$ is a $[6,4]$-graph, every vertex $b \in B$ has at least four neighbours in $A$; otherwise the six vertices $A \cup \{b\}$ would induce at most three edges. Also, since $A$ is independent and $\delta(G) \ge 4$, every vertex of $A$ has at least four neighbours in $B$.

For a vertex $b \in B$, define its type $\tau(b)$ as follows. If $b$ is nonadjacent to exactly one vertex of $A$, then $\tau(b)$ is this unique non-neighbour; if $b$ is adjacent to all vertices of $A$, then set $\tau(b) = *$. For $x, y \in B$, let

$$L(x, y) = N_A(x) \cap N_A(y).$$

Then $|L(x, y)| \geq 3$ for all distinct $x, y \in B$. Since $m \geq 6$ and every six vertices of $B$ induce at least four edges, the graph $G[B]$ contains an edge; fix such an edge $uv$.

We shall use the following elementary form of Hall's theorem. Let $A$ be a set of size five and let $L_1, \ldots, L_r$ be subsets of $A$, each of size at least three. If $r \leq 3$, then $L_1, \ldots, L_r$ have distinct representatives. If $r = 4$, then Hall's condition can fail only when all four lists have the same three-element union, equivalently all four lists are equal to one common 3-subset of $A$. If $r = 5$, then Hall's condition can fail only in one of two ways: either some element of $A$ is absent from all five lists, or some four of the lists are equal to one common 3-subset of $A$. This follows directly from Hall's condition, because no subfamily of at most three lists can have union smaller than its size.

We first find cycles of all lengths from 3 to 10. The edge $uv$, together with any vertex of $L(u, v)$, gives a triangle. Also, any two distinct vertices $x, y \in B$ and any two distinct vertices of $L(x, y)$ give a cycle of length four.

We next construct longer small cycles. Suppose that $X \subseteq B$ has size $r \in \{3, 4, 5\}$, contains $u$ and $v$, and is cyclically ordered as $x_1, x_2, \ldots, x_r, x_1$ with $x_1 = u$ and $x_2 = v$. If the sets

$$L(x_1, x_2),\ L(x_2, x_3),\ \ldots,\ L(x_r, x_1)$$

have distinct representatives in $A$, then these representatives, inserted between consecutive vertices of $X$, give a cycle of length $2r$. Since $uv \in E(G)$, replacing the two-edge segment through the representative of $L(u, v)$ by the edge $uv$ gives a cycle of length $2r - 1$.

For $r = 3$, choose any vertex $w \in B \setminus \{u, v\}$ and take $X = \{u, v, w\}$. The Hall observation above gives distinct representatives, because there are only three lists. Thus $G$ contains cycles of lengths five and six.

For $r = 4$, choose two further vertices $w, z \in B \setminus \{u, v\}$. Consider a cyclic ordering in which $u$ and $v$ are consecutive. By the Hall observation, Hall's condition can fail only if the four lists are all equal to the same three-element set. In terms of types, this means that all four vertices have two non-$*$ types and that these two types alternate around the cycle: for an edge $xy$ of the cycle, the list $L(x, y)$ misses exactly the non-$*$ types occurring at $x$ and $y$. If failure occurs for one ordering, then interchanging $w$ and $z$ breaks this alternation while keeping $u$ and $v$ consecutive. Hence the required distinct representatives exist for one of the two orderings, and $G$ contains cycles of lengths seven and eight.

It remains to deal with $r = 5$. For $a_i \in A$, let

$$B_i = \{b \in B : ba_i \notin E(G)\}.$$

The sets $B_i$ are pairwise disjoint, and $|B_i| \leq m - 4$ for every $i$, because $a_i$ has at least four neighbours in $B$. We choose three vertices from $B \setminus \{u, v\}$ so that, in the resulting five-set $X$, no type from $A$ occurs more than three times. This is always possible. Indeed, if $u$ and $v$ have the same non-$*$ type $a_i$, then $|B \setminus B_i| \geq 4$, so we can choose the three vertices outside $B_i$. Otherwise, let $I$ be the set of indices $i$ for which one of $u$ and $v$ lies in $B_i$. Then $|I| \leq 2$,

and no set $B_i$ with $i \in I$ contains both $u$ and $v$. From $B \setminus \{u, v\}$ delete all but two vertices of each $B_i$ with $i \in I$. If one class is truncated, then at least

$$m - 2 - (|B_i| - 3) = m + 1 - |B_i| \geq 5$$

vertices remain. If two classes are truncated, say $B_i$ and $B_j$, then at least

$$m - 2 - (|B_i| - 3) - (|B_j| - 3) = m + 4 - |B_i| - |B_j| \geq 4$$

vertices remain, since the classes $B_i$ are pairwise disjoint. If no class is truncated, then at least $m - 2 \geq 4$ vertices remain. In all cases, at least three vertices remain, and choosing any three of them gives the desired set $X$.

Choose a cyclic ordering of $X$ in which $u$ and $v$ are consecutive and, whenever some non-$*$ type occurs three times in $X$, the two vertices of $X$ not of this type are consecutive. Such an ordering exists as follows. If no type occurs three times, any cyclic ordering with $u$ and $v$ consecutive will do. If a type occurs three times, then it is the only type with this property; order the five vertices so that the two vertices not of this type are consecutive, and place the consecutive pair $u, v$ consecutively as well. These two adjacency requirements are compatible even when the two pairs overlap.

We claim that the five lists determined by this ordering satisfy Hall's condition. By the Hall observation, there are only two possible failures. First, suppose that some vertex $a \in A$ is absent from all five lists. Then every edge of the five-cycle has at least one endpoint of type $a$. Thus the vertices of type $a$ form a vertex cover of a 5-cycle. Since no type occurs more than three times, type $a$ occurs exactly three times, and the two vertices not of type $a$ are nonadjacent on the chosen cycle. This contradicts the choice of the ordering. Second, suppose that four lists are equal to the same three-element set. The corresponding four edges form a path on the five vertices, and along this path the endpoint types must alternate between two non-$*$ types. Hence one of the two types occurs three times and the two vertices not of this type are nonadjacent in the five-cycle, again contradicting the choice of the ordering. Therefore Hall's condition holds, the required representatives exist, and $G$ contains cycles of lengths nine and ten.

Thus $G$ contains cycles of all lengths from 3 to 10.

Now let $\ell$ be an integer with $11 \leq \ell \leq n$, and put $s = \ell - 5$. Then $6 \leq s \leq m$. We shall choose a set $S \subseteq B$ with $|S| = s$ such that

$$|S \cap B_i| \leq s - 3 \qquad \text{for every } i \in \{1, 2, 3, 4, 5\}. \tag{9}$$

For this, set

$$r_i = \max\{0, |B_i| - (s - 3)\}.$$

It is enough to delete at least $r_i$ vertices from each $B_i$. Since the sets $B_i$ are pairwise disjoint, we have $\sum_i r_i \leq m - s$. Indeed, if only one $r_i$ is positive, then

$$r_i \leq (m - 4) - (s - 3) = m - s - 1 \leq m - s.$$

If $h \geq 2$ of the numbers $r_i$ are positive, then

$$\sum_{i=1}^{5} r_i \leq m - h(s - 3) \leq m - s,$$

because $h(s-3) \geq s$ for $s \geq 6$. Hence we may choose a set $T \subseteq B$ of size $m-s$ containing at least $r_i$ vertices from each $B_i$, and set $S = B \setminus T$. Then $S$ satisfies (9).

Let

$$H = G[A \cup S].$$

The graph $H$ is an induced $[6,4]$-subgraph of $G$ of order $\ell$. Every vertex of $S$ has at least four neighbours in $A$, and by (9) every vertex $a_i \in A$ has at least three neighbours in $S$. Thus

$$\delta(H) \geq 3.$$

We claim that $H$ is 2-connected. It suffices to show that $H - z$ is connected for every vertex $z \in V(H)$. If $z \in A$, then every vertex of $S$ has at least three neighbours in $A \setminus \{z\}$, and every vertex of $A \setminus \{z\}$ has a neighbour in $S$. A component of $H - z$ meeting $S$ therefore contains at least three vertices of $A \setminus \{z\}$, so two such components cannot be disjoint; also no component is contained entirely in $A \setminus \{z\}$. Hence $H - z$ is connected. If $z \in S$, then every vertex of $S \setminus \{z\}$ has at least four neighbours in $A$, and every vertex of $A$ has at least two neighbours in $S \setminus \{z\}$. A component of $H - z$ meeting $S \setminus \{z\}$ therefore contains at least four vertices of $A$, so two such components cannot be disjoint; again no component is contained entirely in $A$. Hence $H - z$ is connected. Thus $H$ is 2-connected.

Since $H$ is a 2-connected $[6,4]$-graph of order $\ell \geq 10$ with $\delta(H) \geq 3$, Theorem 1.6 implies that $H$ is hamiltonian. Therefore $G$ contains a cycle of length $\ell$.

As $\ell$ was arbitrary in $\{11, 12, \ldots, n\}$, and we already found cycles of lengths $3, 4, \ldots, 10$, the graph $G$ is pancyclic. □

*Proof of Theorem 1.5.* Let $G$ be a 2-connected $[6,4]$-graph of order $n \geq 153$ with $\delta(G) \geq 4$. By Theorem 1.6, the graph $G$ is hamiltonian. Since $G$ is a $[6,4]$-graph, it contains no independent set of order six; hence $\alpha(G) \leq 5$.

If $\alpha(G) = 5$, then $G$ is pancyclic by Corollary 5.6. We may therefore assume that $\alpha(G) \leq 4$. By Theorem 5.4 with $k = 4$, the graph $G$ contains cycles of every length from

$$2 \cdot 4^2 + 2 \cdot 4 = 40$$

to $n$. It remains to find cycles of lengths $3, 4, \ldots, 39$.

Since $\alpha(G) \leq 4$, the graph $G$ has no independent set of order five. Thus $G$ contains a cycle $C_\ell$ whenever $n \geq r(C_\ell, K_5)$. For $\ell = 3, 4$, the upper bound in Theorem 5.5 gives

$$r(C_3, K_5) \leq 32 < 153 \leq n, \qquad r(C_4, K_5) \leq 60 < 153 \leq n.$$

For every $5 \leq \ell \leq 39$, the exact part of Theorem 5.5 gives

$$r(C_\ell, K_5) = 4(\ell - 1) + 1 \leq 4 \cdot 38 + 1 = 153 \leq n.$$

Hence $G$ contains cycles of all lengths from 3 to 39.

Combining these cycles with the cycles of all lengths from 40 to $n$, we conclude that $G$ is pancyclic. □

Consequently, the $p = 4$ case of Conjecture 1.3 holds for all graphs of order at least 153. Thus any possible counterexample to the $p = 4$ case of Zhan's conjecture must have order between 11 and 152, and must have independence number three or four.

## 6 Concluding remarks

We constructed a counterexample showing that the case $p = 5$ of Zhan's pancyclicity conjecture does not hold. For $p = 4$, we proved that every 2-connected $[6, 4]$-graph of order at least 153 and with minimum degree at least 4 is pancyclic. Thus the only remaining part of the $p = 4$ case lies in the finite range $11 \leq n \leq 152$, and any counterexample in this range must have independence number three or four.

Professor Xingzhi Zhan posed the following conjecture, which strengths Theorems 1.4 and 1.5.

**Conjecture 6.1** (Zhan, September 2025 [27]). *Let $p \in \{3, 4\}$. Every 2-connected $[p + 2, p]$-graph of order at least $2p + 3$ and with minimum degree at least $p$ contains a pancyclic edge.*

The $p = 2$ case of this conjecture was proved by Li and Zhan [18]. In the same paper, in order to further investigate the structure of pancyclic edges, they asked whether there exists a positive integer $k$ such that every $k$-connected vertex-pancyclic graph contains a pancyclic edge. Very recently, this question was answered in the negative by Xu and Zhou [24]. Our present methods do not prove the pancyclic edge version for $[6, 4]$-graphs. However, in the terminology of $A$-cyclic edges introduced in [18], Theorem 5.4 gives a partial long-cycle conclusion: if $G$ is hamiltonian and $\alpha(G) \leq k$, then every edge of a Hamilton cycle is $\{2k^2+2k, 2k^2+2k+1, \ldots, n\}$-cyclic. Hence every graph covered by Theorem 1.5 contains a $\{60, 61, \ldots, n\}$-cyclic edge, and in the subcase $\alpha(G) \leq 4$ it contains a $\{40, 41, \ldots, n\}$-cyclic edge.

## Declaration

**Conflict of interest.** The authors declare that there is no conflict of interest.

**Data availability.** Data sharing is not applicable to this article as no datasets were generated or analysed.

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